



Figure 2: PaaS reference architecture

change dynamically because of auto-scaling, failures and upgrades. Consequently, client code needs to use a more elaborate service discovery mechanism to identify the available instance.

Configuration: This externalises application configuration to a centralised repository across all the environments. It decouples the configuration from the implementation and it is possible to update the configuration without impacting any of the applications. Each update on the configuration files in the repository is automatically propagated to the running application instances.

Processes and integration: This involves transversal solutions, which facilitate the implementation of business processes and integration between the various services making up the system.

Messaging: Provides stable communication patterns that scale across multiple network topologies. It supports communication via point-to-point or the pub sub-model. Applications are likely to be written in multiple programming languages, so it is necessary that the communication be 'polyglot' and therefore data-centric. This provides maintainability, scalability, efficiency and robustness for applications.

Logging and metrics: This is a system for aggregating and streaming logs and metrics from all applications. Metrics are important to measure and monitor an application or a service's health and performance. A few critical metrics to be captured are: service/system availability, reliability, response time, throughput/bandwidth, capacity, scalability, latency, etc. Also, logs from all the applications will be aggregated and can be used or archived by third-party applications for further analysis.

Security: This is responsible for authentication and securing the service endpoints. Other aspects like the access control and encryption types used are also covered.

Application framework: This provides various platform runtimes for cloud native applications to execute, as well as visual assembly capability to assemble an application on the cloud.

Data layer: This offers flexible and scalable information storage, together with the tools necessary to exploit and obtain the maximum value from data.

IaaS: This provides self-service for infrastructure elements which sustain the application's components.

Containers as a Service: This allows users to have a massively scalable, easily automated and replicated execution environment.

Zero downtime deployments: This is about frequently deploying a new version of a service without impacting its availability. Multiple configurations need to be done in PaaS to achieve zero downtime deployments.

Failover and recovery: This deals with failover, disaster recovery strategies and the configurations to be implemented.

Scaling: This is responsible for scaling the application(s) when required. It deals with scaling strategies and configurations.

Security patching: This is responsible for applying security patches to various components of the platform.

Platform upgrades: This is responsible for applying product upgrade patches as required.

Cloud services: Any external dependency that the cloud native application has is a backing service. Cloud services will be plugged to the cloud native application using the PaaS application framework. User defined services can also be plugged to the cloud native application as cloud services. This might depend on the PaaS platform in question. The categories of cloud services are databases, analytics, Big Data, object stores, etc.

Operation: This ensures that the system can be monitored and managed in a simple way by the operations team.

Key considerations when selecting open source PaaS platforms

The key factors to be taken in consideration when choosing a PaaS platform are listed below:

- Deployment flexibility
- Ease of operations
- Choice of application stacks
- Language, database and framework support
- Scaling capabilities
- QoS
- Tooling for development and operations
- How well it meets the needs of the business

Key open source PaaS platforms

- a) **Cloud Foundry:** This is an open Platform as a Service that provides a choice of clouds, developer frameworks and application services. Cloud Foundry makes it faster and easier to build, test, deploy and scale applications. It has different distributions, of which Pivotal and IBM are popular. It contains the application runtime and container runtime. It also has the Pivotal application and container services.
- b) **OpenShift:** This is Red Hat's cloud computing PaaS offering. It is an application platform in the cloud where developers and teams can build, test, deploy and run their apps.